IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MATHEMATICS FOR ENGINEERS IIA

Tuesday, 7 January 2025, 10:00 AM

Time allowed: 1:30 hours

There are TWO questions on this paper.

Answer ALL questions.

Question ONE is worth 20 marks. Question TWO is worth 30 marks.

NO CALCULATORS ALLOWED

Any special instructions for invigilators and information for candidates are on page 1.

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ELEC50011A - MATHEMATICS 2A

The residue of a complex function $F(z)$ at a pole $z = a$ of multiplicity m is given by the expression

$$\lim_{z\to a}\left\{\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left[(z-a)^m F(z)\right]\right\}.$$

1. a) Let a mapping from the $z-$plane to the $w-$ plane be given by

$$w(z) = \frac{z-1}{z+i}.$$

Determine the image in the $w-$plane of the $x-$axis in the $z-$plane. Which curve in the $z-$plane maps to the line $u = v$ in the $w-$plane? . [5]

b) Show that the real integral

$$\int_0^{2\pi} e^{\cos\theta}\,\cos(\sin\theta)\,d\theta \qquad (1)$$

can be written in terms of the imaginary part of the complex contour integral

$$\oint_C \frac{e^z}{z}\,dz \qquad (2)$$

where C is the unit circle in the complex plane. Use Cauchy's residue theorem to evaluate (2), and hence obtain (1). [6]

c) To obtain the value of the real, improper integral

$$I = \int_{-\infty}^{+\infty} \frac{\cos x}{(x^2+4)^2}\,dx,$$

we proceed as follows:

(i) The function $F(z) = \dfrac{e^{iz}}{(z^2+4)^2}$ has two double poles at $z = \pm 2i$. Obtain the residue of $F(z)$ at the pole in the upper half-plane. [3]

(ii) Choose an appropriate semi-circular contour C lying on the real axis and in the upper half-plane to evaluate the complex integral

$$\oint_C F(z)\,dz.$$

Hence, or otherwise, find the value of I. Be sure to explain your reasoning! [6]

2. a) A harmonic function is given as $v(x,y) = \cos y \cosh x + 2 \sin y \sinh x$. Find its harmonic conjugate $u(x,y)$, and hence the analytic function $f(z)$ where $f(z) = u + iv$. [4]

b) An invertible matrix is given as

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} .$$

Use Gram-Schmidt orthogonalization and the columns of A to find an orthonormal basis for $\mathbb{R}^3$. Use the basis to find an orthogonal matrix Q and an upper-triangular matrix R, so that $A = QR$: the $QR-$factorization of A. [6]

c) Find an orthogonal matrix S and a diagonal matrix Λ, so that $A = S\Lambda S^T$: the orthogonal diagonalization of the 3×3 symmetric matrix [6]

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} .$$

d) Let λ, γ be distinct eigenvalues of the symmetric matrix A, with respective eigenvectors $\underline{\mathbf{u}}, \underline{\mathbf{v}}$. Prove that $\underline{\mathbf{u}}, \underline{\mathbf{v}}$ are orthogonal vectors. [4]

e) Given

$$A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \\ 2 & 2 \end{pmatrix} ,$$

its Singular Value Decomposition (SVD) is $A = U\Sigma V^T$.

Explain *in words* how the first stage of finding the SVD of A involves the orthogonal diagonalization of a symmetric matrix. Finally, find matrices U, Σ and V, giving the SVD of A. [10]